반도체공정

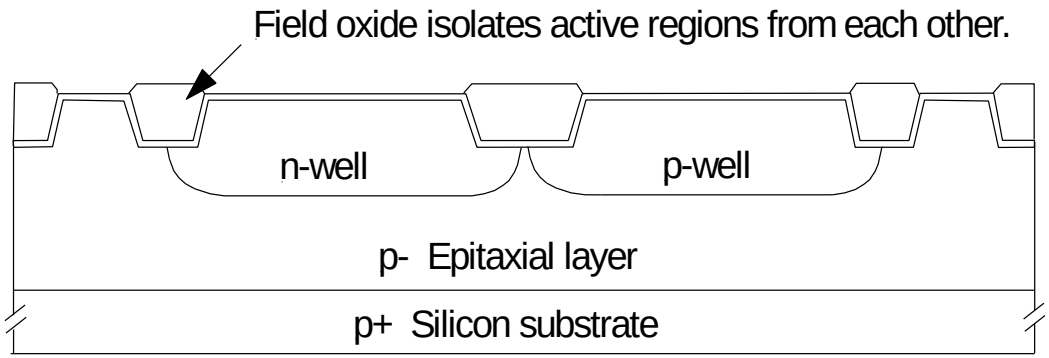
#6 산화공정

산화공정

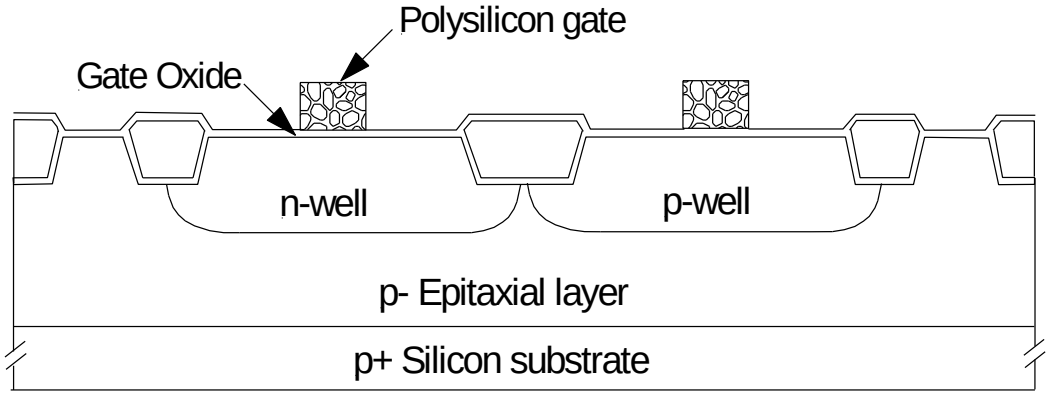
- 실리콘 웨이퍼 위에 실리콘 다이옥사이드 (SiO_2)를 성장시키는 공정
- 용도
  - 게이트 산화층 (gate oxide) √
  - 절연층 (field oxide) √
  - 캐퍼시터 (capacitor)
  - 마스크 (mask) √
  - 표면 보호막 (passivation)
  - 소자간 격리 (isolation)

산화물 용도

- Field Oxide Layer

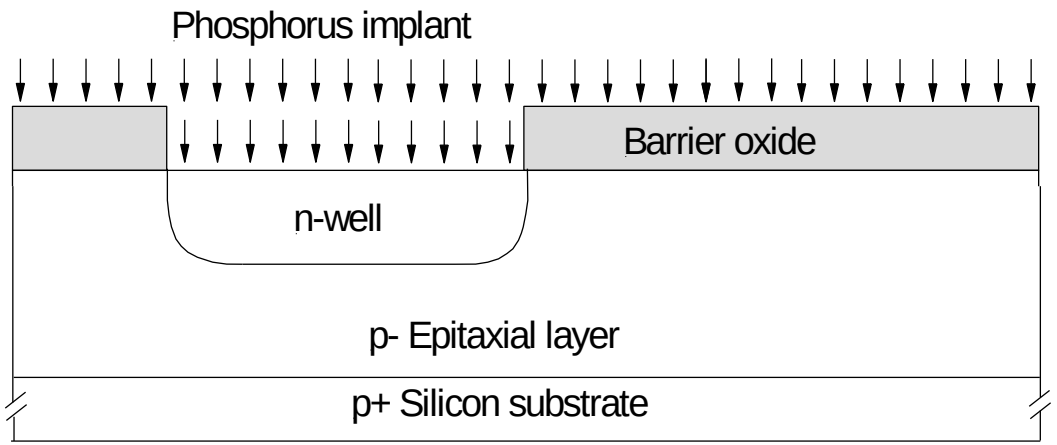


- Gate Oxide Dielectric



산화물 용도

- Oxide Layer Dopant Barrier – 마스크 (mask)



산화공정

- 산화공정의 종류
  - 열 산화
    - Dry oxidation : $Si + O_2 \rightarrow SiO_2$
    - Wet oxidation : $Si + 2H_2O \rightarrow SiO_2 + 2H_2$
  - SiO_2 증착
    - PVD : sputtering, evaporation
    - CVD : PECVD, Thermal CVD
  - 양극산화 (anodic oxidation)

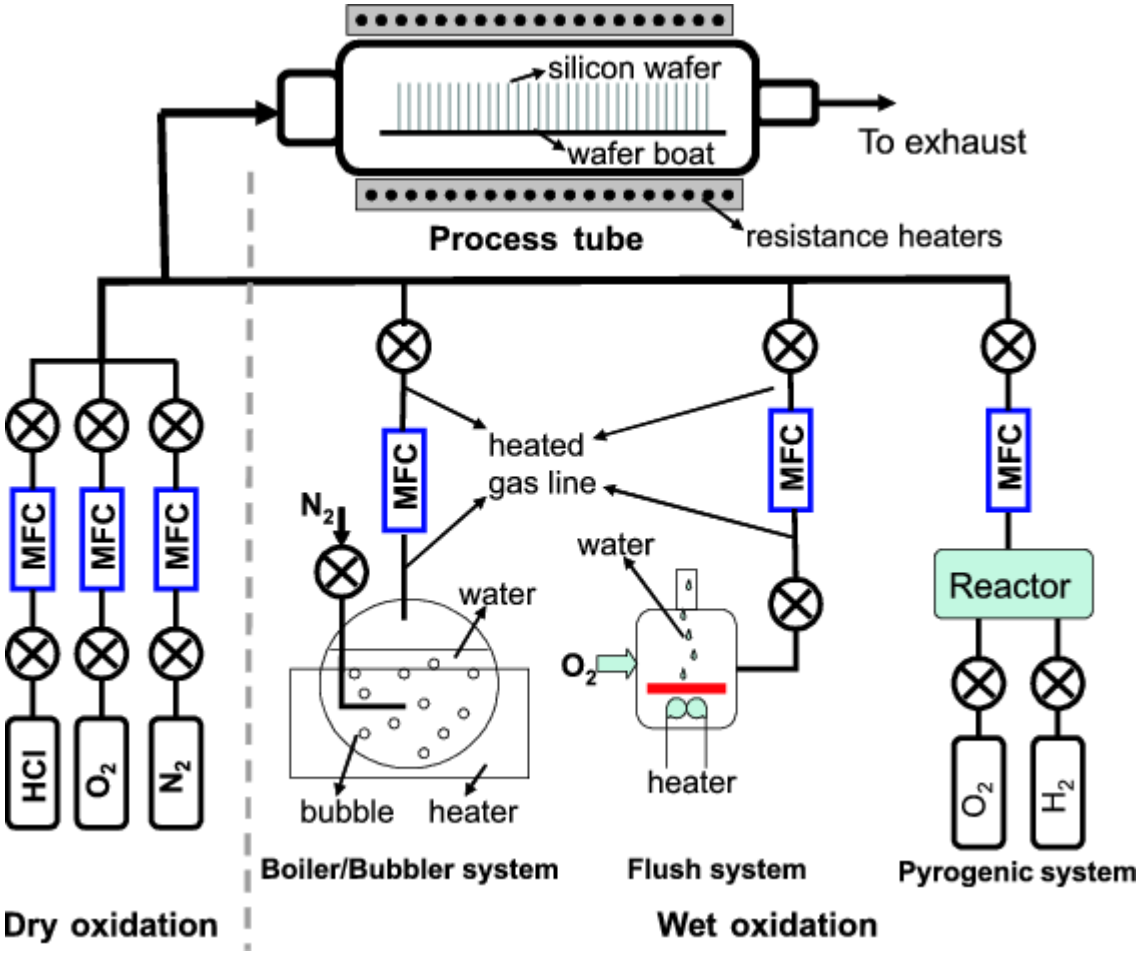
산화공정

- 습식산화공정 vs 건식산화공정

	장점	단점
습식산화공정	산화속도가 건식에 비하여 빠르다.	절연전압이 건식에 비하여 낮다.
건식산화공정	습식에 비하여 절연전압이 높다. (5~10MV/cm)	성장속도가 느리다.

산화공정

- 열산화공정 모식도

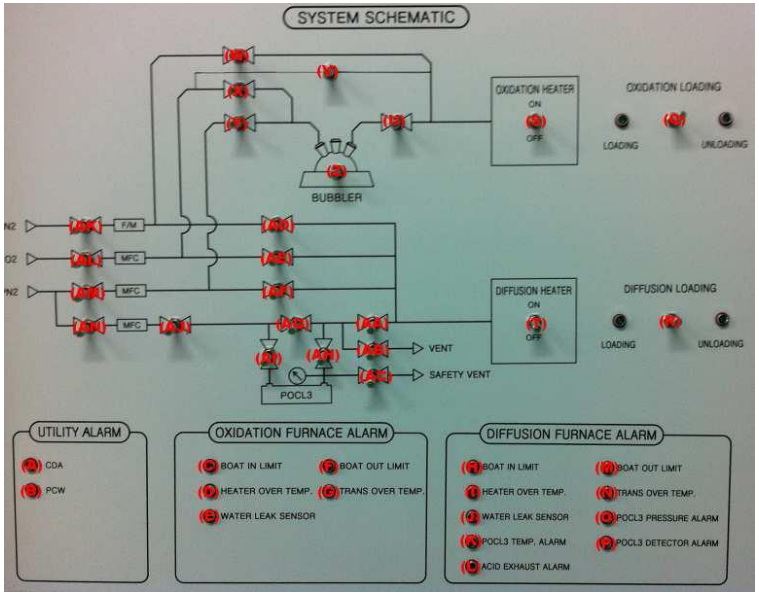


열산화공정

- Oxidation Furnace

열산화공정

- 컨트롤 패널 & 시스템 스케메틱



열산화공정

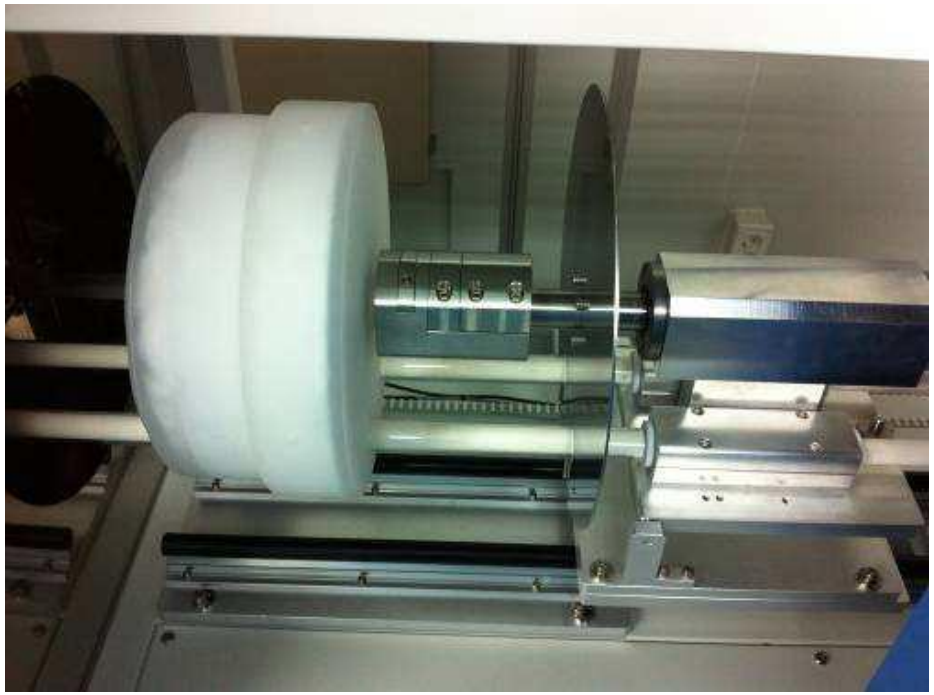
- 전원부

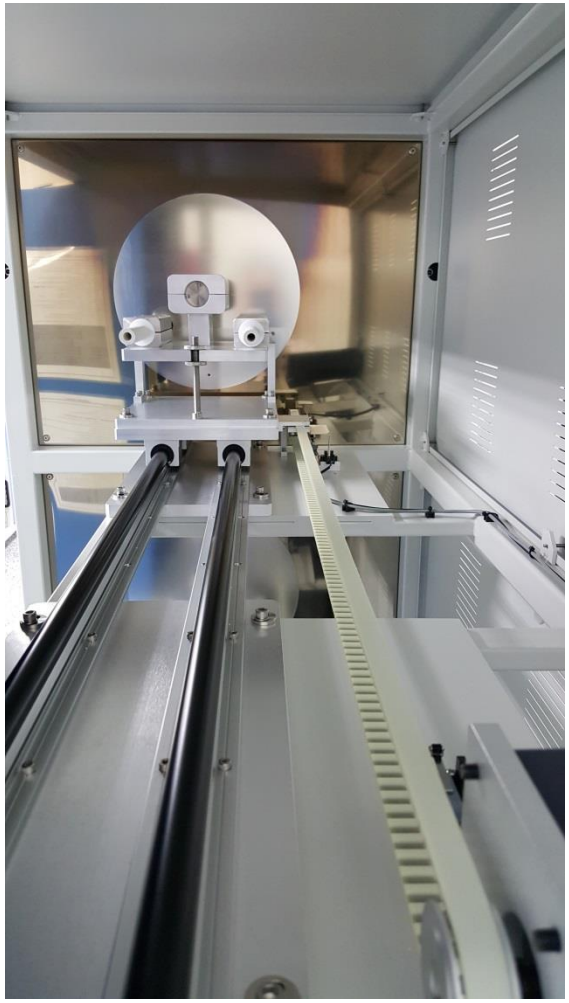
HEATER#1
(OXIDATION)
HEATER#2
(DIFFUSION)
MAIN
ON
OFF
ON

SYSTEM POWER
ON
OFF

열산화공정

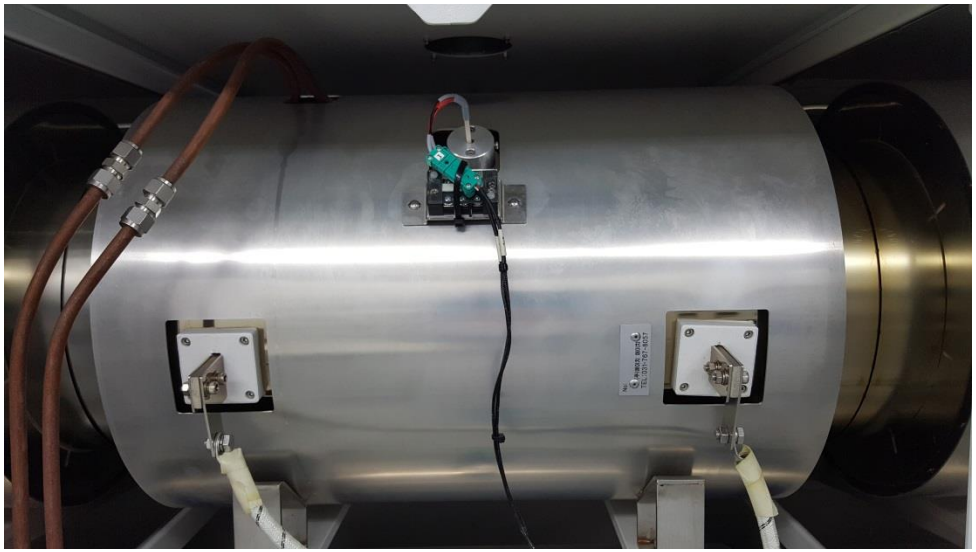
- 로더모듈





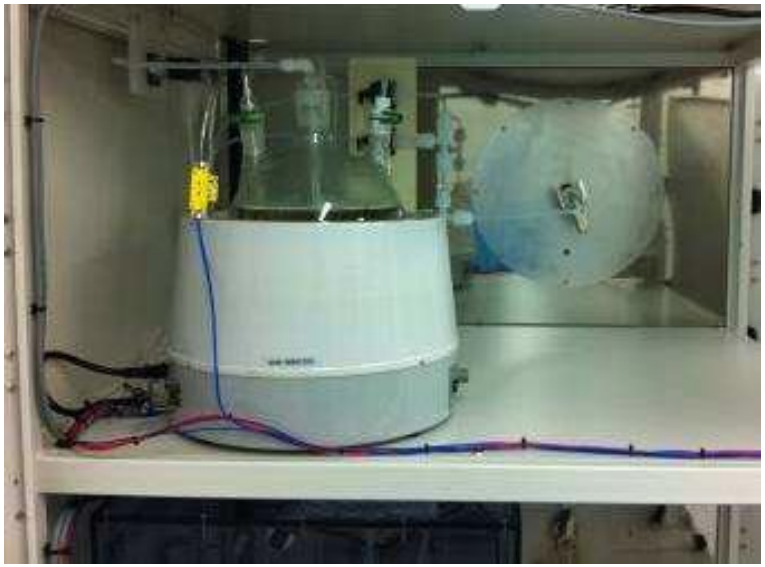
열산화공정

- 히터모듈



열산화공정

- 가스모듈



열산화공정

- Wet Scrubber